

EXPERIMENT 8

```
clc;
```

```
clear;
```

```
close all;
```

```
% Steering Angles (degrees)
```

```
angle = [-60 -30 0 30 60];
```

```
% Normalized Gain
```

```
gain = ones(size(angle));
```

```
% Create Polar Plot
```

```
figure;
```

```
polarplot(deg2rad(angle), gain, 'o-', 'LineWidth', 2);
```

```
% Add Title
```

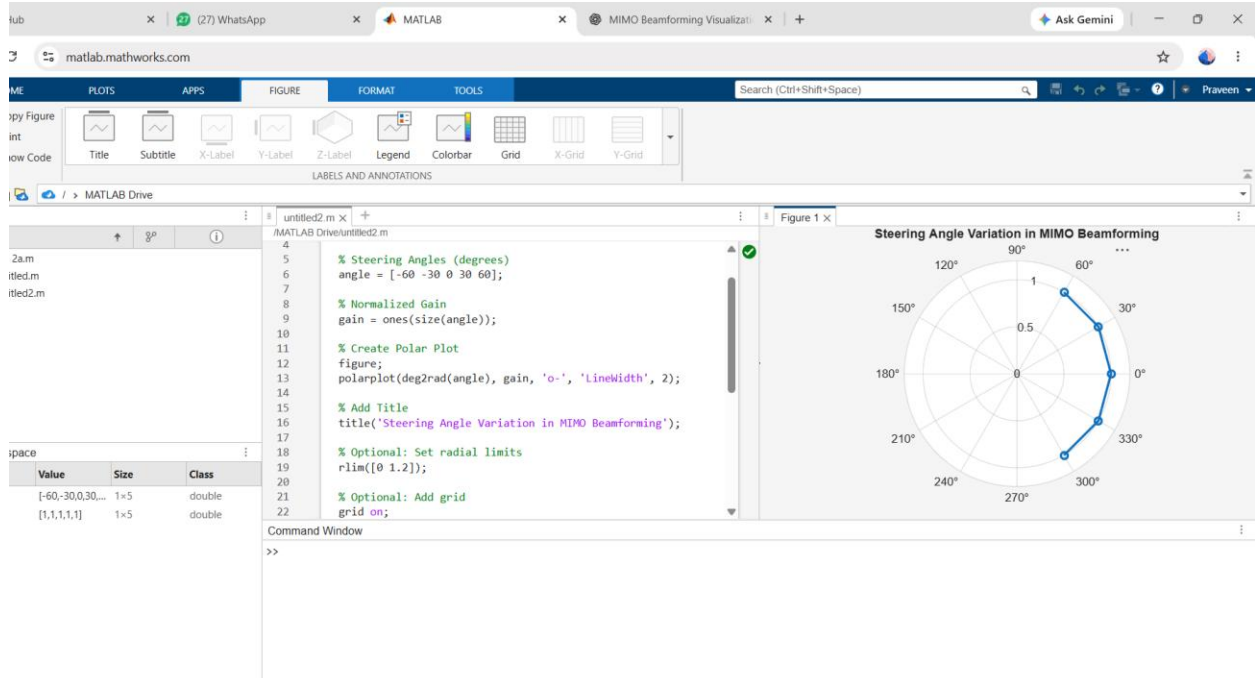
```
title('Steering Angle Variation in MIMO Beamforming');
```

```
% Optional: Set radial limits
```

```
rlim([0 1.2]);
```

```
% Optional: Add grid
```

```
grid on;
```



OBJ 1

clc;

clear;

close all;

% Steering Angles (degrees)

angle = [-60 -30 0 30 60];

% Array Gain (dB)

gain = [9 12 15 12 9];

% Create Bar Graph

figure;

bar(angle, gain);

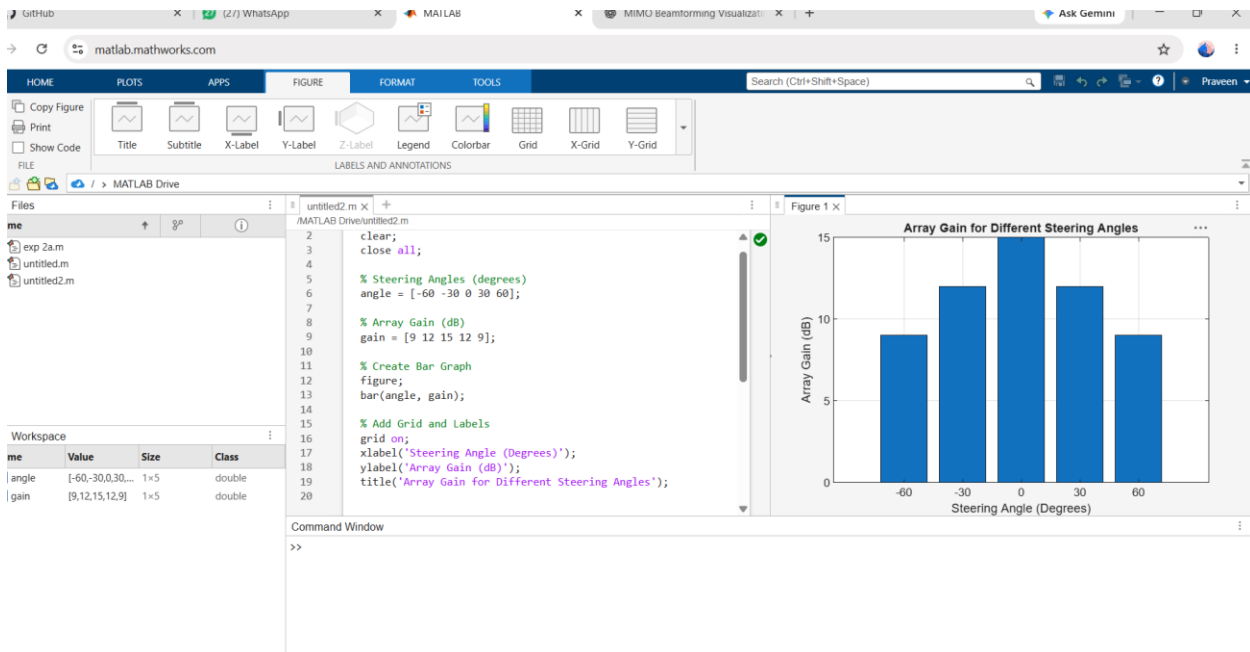
% Add Grid and Labels

grid on;

xlabel('Steering Angle (Degrees)');

ylabel('Array Gain (dB)');

title('Array Gain for Different Steering Angles');



OBJ 2

clc;

clear;

close all;

% Desired Steering Angles (degrees)

desired = [-60 -30 0 30 60];

% Actual Beam Directions (degrees)

actual = [-58 -32 1 29 62];

% Beam Direction Error (Absolute Error)

error = abs(desired - actual);

% Plot Beam Direction Error

figure;

plot(desired, error, 'o-', 'LineWidth', 2);

grid on;

xlabel('Desired Steering Angle (Degrees)');

ylabel('Beam Direction Error (Degrees)');

title('Beam Direction Error for Different Steering Angles');

